Elif Uskuplu

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Research Interests

Homotopy type theory and its extensions; formalized mathematics and proof assistants; categorical semantics of type theories; natural logic; ethical use of AI in mathematics and education; combinatorial design theory and its applications in coding theory.

Education

2023 Ph.D. in Mathematics, University of Southern California
Dissertation: *Formalizing two-level type theory with cofibrant exo-nat*
Advisors: Aaron Lauda (USC) and Michael Shulman (University of San Diego)

2019 M.S. in Mathematics, Koç University
Advisor: E. Şule Yazıcı

2017 B.Sc. in Mathematics, Istanbul Bilgi University

Employment

2024 – present Zorn Postdoctoral Fellow, Department of Mathematics, Indiana University Bloomington
Mentor: Lawrence S. Moss

2025 – 2026 Lean Expert (part-time), Vals AI
Contributed to FormalProofBench, a benchmark for formal theorem proving in Lean 4.

Spring 2024 Lecturer, Department of Mathematics, University of Southern California

Honors, Awards and Grants

2027 Research membership, AxIOM: Building the Mathematical Library of the Future, SLMath, Berkeley (March–April 2027)

2026 Max Zorn Teaching Award, Department of Mathematics, Indiana University

2025 Travel and participation grant, Simons Foundation, MPS Workshop on Lean

2023 Edward and Dolores Blum Research Prize, Department of Mathematics, USC

2018 Outstanding Teaching Award, Koç University

2017 – 2018 TÜBİTAK ARDEB 1001 project scholarship, grant no. 116F166 (PI: E. Şule Yazıcı)

2016 – 2017 Cahit Arf Scholarship, Turkish Mathematical Society

2013 – 2017 Board of Trustees Scholarship, Istanbul Bilgi University

2012 – 2017 State scholarship, Higher Education Credit and Hostels Institution (KYK)

Graduate Research Support

2021 – 2023 AFOSR award FA9550-21-1-0009 (PI: Michael Shulman)

2019 – 2021 NSF DMS-1902092, ARO W911NF-20-1-0075, and Simons Foundation (PI: Aaron Lauda)

Publications

Journal Articles

7. **Elif Uskuplu**. *Formalizing two-level type theory with cofibrant exo-nat*. Mathematical Structures in Computer Science 35 (2025), e30. doi:10.1017/S0960129525100297
6. Diane Donovan, Asha Rao, **Elif Uskuplu**, E. Şule Yazıcı. *QC-LDPC codes from difference matrices and difference covering arrays*. IEEE Access 11 (2023), 52141–52157. doi:10.1109/ACCESS.2023.3279327
5. Diane Donovan, Aiden Price, Asha Rao, **Elif Uskuplu**, E. Şule Yazıcı. *High-rate LDPC codes from partially balanced incomplete block designs*. Journal of Algebraic Combinatorics 55 (2022), 259–275. doi:10.1007/s10801-021-01111-0

Book Chapters

4. Lawrence S. Moss, **Elif Uskuplu**. *Prolegomena to semantic and algorithmic treatments of polarity tagging*. To appear in an edited volume.

Conference Proceedings

3. Eric Jonathan Wang, **Elif Uskuplu**. *Formalizing the Bruck–Ryser–Chowla Theorem: Combinatorial Design Theory in Lean*. 17th International Conference on Interactive Theorem Proving (ITP 2026), LIPIcs 382 (2026), Paper 19. doi:10.4230/LIPIcs.ITP.2026.19

Workshop Papers and Preprints

2. Nikil Ravi, Kexing Ying, Vasilii Nesterov, Rayan Krishnan, **Elif Uskuplu**, Bingyu Xia, Janitha Aswedige, Langston Nashold. *FormalProofBench: Can Models Write Graduate Level Math Proofs That Are Formally Verified?* ICLR 2026 Workshop VerifAI-2 (2026). arXiv:2603.26996
1. **Elif Uskuplu**, Lawrence S. Moss, Valeria de Paiva. *KnowTeX: Visualizing Mathematical Dependencies*. Preprint (2026). arXiv:2601.15294

In Preparation

- Lawrence S. Moss, **Elif Uskuplu**. *A Semantic Treatment of Polarity Tagging*.
- **Elif Uskuplu**, Lawrence S. Moss. *An Algorithmic Treatment of Polarity Tagging*.

Conference Abstracts

- Valeria de Paiva, Lawrence S. Moss, **Elif Uskuplu**. *Enhancing Dependency Graphs for Research and Alignment in Math*. Math Datasets and Libraries: Alignments and Comparisons (ALIGN 2025), 2025.
- Thorsten Altenkirch, Ambrus Kaposi, Michael Shulman, **Elif Uskuplu**. *From parametricity to identity types*. 31st International Conference on Types for Proofs and Programs (TYPES 2025), 2025.
- Thorsten Altenkirch, Ambrus Kaposi, Michael Shulman, **Elif Uskuplu**. *Internal relational parametricity, without an interval*. 30th International Conference on Types for Proofs and Programs (TYPES 2024), 2024.
- **Elif Uskuplu**. *Cofibrancy of the exo-type of natural numbers*. Second International Conference on Homotopy Type Theory (HoTT 2023), 2023. Abstract
- **Elif Uskuplu**. *An experiment with 2LTT features in Agda*. Workshop on Homotopy Type Theory/Univalent Foundations (HoTT/UF 2022), 2022. Abstract

Software and Formalization Projects

9. agda-unimath: univalent mathematics in Agda. Maintainer.
8. KnowTeX: visualizing mathematical dependencies.
7. Polarity Tagger: a web tool that parses English sentences and computes monotonicity polarities for natural logic.
6. Combinatorial design theory in Lean 4, with Eric Jonathan Wang.

5. Two-level type theory in Agda: documentation for the two-level flag, with Agda contributors.
4. 2LTT and the Univalence Principle in Agda.
3. Narya: a proof assistant for higher-dimensional type theory. Contributor.
2. PrimeNumberTheoremAnd: formalization of the Prime Number Theorem in Lean 4. Contributor.
1. Simply typed lambda calculus with de Bruijn indices in Lean 4.

Talks

Oct 2025	*Enhancing Dependency Graphs for Research and Alignment in Math*. Math Datasets and Libraries: Alignments and Comparisons (ALIGN 2025).
May 2025	*Monotonicity and Natural Logic: Tracking Meaning with Arrows*. AWM Research Symposium, Special Session on Logic Across Mathematics.
Jan 2025	*Formalizing locally nameless syntax with cofinite quantification*. Workshop on the Implementation of Type Systems (WITS 2025), co-located with POPL 2025.
Sep–Oct 2024	*From Propositions to Proof Assistants: A Journey through Type Theory* (two talks). Logic Seminar, Indiana University Bloomington.
Feb 2024	*Navigating the New Frontier: Formalized Mathematics*. Women in Mathematics in Southern California (WiMSoCal 2024).
May 2023	*Cofibrancy of the Exo-type of Natural Numbers*. Second International Conference on Homotopy Type Theory (HoTT 2023), Carnegie Mellon University.
Dec 2022	*Homotopy Type Theory vs Set Theory*. Research Seminar, Galatasaray University.
Aug 2022	*Proof Assistants: Agda for Beginners*. SageDays Istanbul.
Jul 2022	*An Experiment with 2LTT Features in Agda*. Workshop on Homotopy Type Theory/Univalent Foundations (HoTT/UF 2022).
Jul 2022	*Formalization of 2LTT in Agda*. MURI Meeting, Carnegie Mellon University.
Feb 2022	*A Brief Introduction to Homotopy Type Theory*. Graduate Colloquium, Department of Mathematics, USC.

Research Programs and Workshops

Aug 2026	*Teaching Higher Category Theory with Computers*, ICERM, Providence
Jun 2025	*MPS Workshop on Lean*, Simons Foundation, New York
Jun 2023	*Summer Graduate School: Formalization of Mathematics*, SLMath, Berkeley
May 2023	*HoTT 2023 and MURI 2023* (joint event), Carnegie Mellon University
Jul–Aug 2022	*HoTTEST Summer School 2022* (online)
Jun 2022	*MURI Meeting 2022*, Carnegie Mellon University
Feb 2022	*Logic and Higher Structures*, CIRM, Marseille

Teaching

Zorn Postdoctoral Fellow, Indiana University Bloomington

Fall 2026	Math 301, Linear Algebra and Applications Math 384, Logic
Spring 2026	Math 782, Selected Topics in Mathematical Logic: Introduction to Type Theory and Proof Assistants (graduate)

Fall 2025	Math 106, The Mathematics of Decision and Beauty Math 384, Logic
Spring 2025	Math 303, Linear Algebra for Undergraduates Math 391, Introduction to Mathematical Reasoning
Fall 2024	Math 211, Calculus I

Lecturer, University of Southern California

Spring 2024	Math 225A, Linear Algebra and Differential Equations

Teaching Assistant

2019 – 2023	University of Southern California: Real Analysis (525A), Topology (540), Fundamental Concepts of Analysis (425A), Algebra (510B), Fundamental Concepts of Modern Algebra (410), Fundamental Principles of Calculus (118), Theory of Numbers (430), Calculus I (125)
2018 – 2019	Istanbul Bilgi University: Topology (245), Linear Algebra (217), Probability and Statistics for Engineers and Scientists (240), Intuitive Set Theory (111), Analytic Geometry and Linear Algebra I (181), Advanced Calculus I (253)
2017 – 2019	Koç University: Finite Mathematics (MATH 101), Mathematics for Everyday Life (EQUR 102), Introduction to Symbolic Logic (EQUR 121)

Mentoring

Summer 2025	REU mentor, Indiana University. Student: Eric Jonathan Wang. The project resulted in the ITP 2026 paper.
Summer 2023	Mentor, Directed Reading Program Türkiye
Spring 2023	Volunteer, Mathematics Solidarity Platform

Professional Service

2026	Program committee member, LSFA 2026: Logical and Semantic Frameworks with Applications
2025	Referee, *Forum of Mathematics, Sigma*
2025	Reviewer, 31st International Conference on Computational Linguistics (COLING 2025)
2022 – 2023	Research event coordinator, USC Women in Mathematics